SX Steady-State Model — Step-by-Step Excel Guide

System: D2EHPA / HCl | **Reference:** Alind Chandra formulation
Solver: TDMA (Thomas Algorithm) implemented as direct Excel formulas
Elements modelled: Pr, Nd, Tb, Dy

What This File Does

This Excel workbook simulates a continuous solvent extraction (SX) circuit at **steady state**. Given a set of operating conditions, it calculates what concentration of each rare earth element (REE) will be found in every mixer-settler stage of the circuit, and where each element ends up — in the raffinate or in the pregnant leach solution (preg).

The workbook has three sheets:

Sheet	Purpose
INPUTS	All parameters you can change
SOLVER	Matrix construction + TDMA solution (do not edit)
RESULTS	Summary table and two charts

Part 1 — Chemistry Background

The Reaction

When rare earth chlorides ($LnCl_3$) in aqueous HCl contact D2EHPA (abbreviated H-DEHP) dissolved in kerosene, the following exchange reaction occurs:

```
3 H-DEHP(org)  +  LnCl₃(aq)  ⇌  Ln-DEHP₃(org)  +  3 HCl(aq)
```

The REE transfers from aqueous to organic; acid is released back into the aqueous phase.

The Distribution Coefficient

The distribution coefficient **D** is defined as:

```
D  =  [Ln in organic phase]  /  [Ln in aqueous phase]
```

For a fixed D2EHPA concentration and temperature, D depends only on the HCl normality:

```
D = P × [HCl]^Q
```

Where **P** and **Q** are element-specific empirical constants determined by shakeout tests. A higher D means the element prefers the organic phase at that acid concentration.

Key insight: **Q is always negative**, so increasing HCl normality drives REEs *back* into the aqueous phase. This is why: - **Extraction** uses low HCl (~0.5 N) → high D → REEs load onto organic - **Scrub** uses moderate HCl (~1.37 N) → lower D → only the most strongly loaded REEs stay - **Strip** uses high HCl (~5 N) → very low D → REEs are forced back to aqueous (preg)

Part 2 — Circuit Topology

The circuit has three sections wired in series. Organic flows in one direction; aqueous flows countercurrent. Here is how the stages are numbered in this model:

```
Stage:     1  2  3 ... 10 | 11 12 ... 20 | 21 22 ... 25
Section:   ←— Extraction —→ ←——— Scrub ———→ ←—— Strip ——→
           (n_ext = 10)      (n_scr = 10)     (n_str = 5)
```

Organic flow direction → (stage 1 to 25, then recycles back)
Aqueous flow direction ← (countercurrent in each section)

Key streams:

Stream	Location	Flow rate
Feed	Aqueous enters at stage **n_ext** (stage 10)	F
Raffinate	Aqueous leaves at stage **1**	S×R + F
Scrub acid	Aqueous enters at stage **n_ext + 1** (stage 11) — as reflux from strip	S×R
Preg	Aqueous leaves at stage **n_ext + n_scr + 1** (stage 21)	S×(1−R)
Strip acid	Aqueous enters at last stage **25**	S
	Organic leaves extraction at stage 10	O

Stream	Location	Flow rate
Loaded organic		

Reflux R is the fraction of the strip section aqueous outflow that is recycled back into the scrub section as scrub solution (instead of exiting as preg).

Part 3 — Mathematical Model

Setting Up the Mass Balances

For each stage, the amount of REE entering equals the amount leaving (steady state). Both organic (y) and aqueous (x) streams carry REE, and they are linked by equilibrium:

```
y_{i,n}  =  k_{i,n} × x_{i,n}
```

where k = D (the distribution coefficient at the HCl normality of that section).

After substituting the equilibrium relation, each stage n gives one equation in terms of the **organic concentrations y only**:

```
a_n × y_{n-1}  +  b_n × y_n  +  c_n × y_{n+1}  =  d_n
```

This is a **tridiagonal equation** — each unknown (y_n) is connected only to its immediate neighbors. The full system for all N stages forms a tridiagonal matrix.

The a, b, c, d Coefficients

The coefficients depend on which section the stage belongs to:

Strip section (stages n_ext+n_scr+1 to N):

```
a_n  =  O                   (organic from previous stage)
b_n  =  −O − S / D_n        (total outflow from stage n)
c_n  =  S / D_{n+1}         (aqueous contribution from next stage)
d_n  =  0
```

Exception: last stage (n = N): `c_N = 0` (no stage beyond N), and d contains strip acid concentration, but since strip acid carries no REE, d_N = 0.

Scrub section (stages n_ext+1 to n_ext+n_scr):

```
a_n  =  O
b_n  =  −O − (S×R) / D_n
c_n  =  (S×R) / D_{n+1}
d_n  =  0
```

Extraction section (stages 1 to n_ext):

```
a_n  =  O    (except a_1 = 0, as no stage precedes stage 1)
b_n  =  −O − (S×R + F) / D_n
c_n  =  (S×R + F) / D_{n+1}    for n < n_ext
c_n  =  (S×R) / D_{n+1}          for n = n_ext (feed cell; F enters here, not carried forw
d_n  =  0    except at the feed cell (n = n_ext):
d_{n_ext}  =  −(Feed_concentration) × F
```

The rightmost column `d` is zero everywhere except the feed stage, which acts as the source term injecting the feed into the system.

Part 4 — Solving with TDMA (Thomas Algorithm)

The tridiagonal system cannot be solved in one step with a direct formula, but it can be solved in two linear sweeps — **no iteration required**.

Step 1 — Forward Elimination (top to bottom)

Compute modified coefficients c′ and d′:

```
c'_1  =  c_1 / b_1
d'_1  =  d_1 / b_1

For n = 2 to N:
  w_n   =  b_n  −  a_n × c'_{n-1}
  c'_n  =  c_n / w_n
  d'_n  =  (d_n  −  a_n × d'_{n-1}) / w_n
```

In the SOLVER sheet, these are the **c′ and d′ columns** for each element. Each row references only the row above — pure top-down calculation.

Step 2 — Back Substitution (bottom to top)

```
y_N  =  d'_N
```

```
For n = N−1 down to 1:
  y_n  =  d'_n  −  c'_n × y_{n+1}
```

In the SOLVER sheet, the **y_org column** for each element references the row *below* it. Excel resolves this automatically through its dependency engine — no circular references.

Getting Aqueous Concentrations

Once y_n (organic) is known, the aqueous concentration follows directly:

```
x_n  =  y_n / D_n
```

These are the **x_Pr, x_Nd, x_Tb, x_Dy columns** (AI–AL) in the SOLVER sheet.

Part 5 — The INPUTS Sheet

Open the INPUTS sheet and change only the **yellow/white cells** in the value columns. Never edit the formula cells in SOLVER or RESULTS.

General Inputs

Cell	Parameter	Description
B5	n_ext	Number of extraction stages
B6	n_scr	Number of scrub stages
B7	n_str	Number of strip stages
B8	HCl Ext [N]	HCl normality in extraction section
B9	HCl Scr [N]	HCl normality in scrub section
B10	HCl Str [N]	HCl normality in strip section
B11	F	Feed flowrate (any consistent units)
B12	S	Strip acid flowrate
B13	O	Organic flowrate
B14	R	Reflux fraction (0.70 = 70%)

> **Important:** If you change n_ext, n_scr, or n_str, the SOLVER sheet rows are fixed at 25 stages. You must keep n_ext + n_scr + n_str = 25.

Distribution Coefficients (D = P × [HCl]^Q)

Rows	Columns	Data
18–21	C	P constant for Pr, Nd, Tb, Dy
18–21	D	Q exponent for Pr, Nd, Tb, Dy

These constants come from shakeout experiments. P sets the overall extractability; Q (always negative) sets how strongly acid suppresses extraction.

Feed Concentrations

Rows	Column	Data
25–28	C	Feed concentration of Pr, Nd, Tb, Dy in g/L

Part 6 — The SOLVER Sheet

Do not edit any cell in this sheet. All cells contain formulas.

Column Organization

Each element (Pr, Nd, Tb, Dy) occupies 8 consecutive columns:

```
Offset 0: D_i      = P × [HCl]^Q  for the section of this stage
Offset 1: a        = sub-diagonal coefficient
Offset 2: b        = main-diagonal coefficient
Offset 3: c        = super-diagonal coefficient
Offset 4: d        = right-hand side (only non-zero at feed stage)
Offset 5: c'       = forward-sweep modified super-diagonal
Offset 6: d'       = forward-sweep modified RHS
Offset 7: y_org    = organic concentration (g/L)  ← SOLUTION
```

Column blocks: - **Pr**: columns C–J - **Nd**: columns K–R - **Tb**: columns S–Z - **Dy**: columns AA–AH

After these blocks: - **AI–AL**: aqueous concentrations x_Pr, x_Nd, x_Tb, x_Dy = y/D - **AM**: total organic g/L (sum of all y) - **AN**: total aqueous g/L (sum of all x)

Row Colors

Color	Section
Yellow	Extraction stages (1–10)

Color	Section
Green	Scrub stages (11–20)
Red/orange	Strip stages (21–25)

How to Read the Results

- **Stage 14 (row 14), column AM**: Loaded organic at extraction exit = total REE on organic entering scrub
- **Stage 1 (row 5), columns AI–AL**: Raffinate concentrations
- **Stage 21 (row 25), columns AI–AL**: Preg concentrations

Part 7 — The RESULTS Sheet

Key Metric (Row 4)

Loaded Organic @ Ext Exit is the total REE loading on the organic leaving the last extraction stage (entering scrub). This is a primary circuit design indicator. Target value for the reference case: **19.05 g/L**.

Compositions & Distributions Table (Rows 8–11)

Column	Meaning
Feed Conc.	Input concentration from INPUTS sheet
Raff Conc.	Aqueous concentration at stage 1 (raffinate)
Preg Conc.	Aqueous concentration at stage 21 (first strip stage = preg exit)
% Feed	Fraction of total feed mass that is this element
% Raffinate	Purity of this element in the raffinate stream
% Preg	Purity of this element in the preg stream

Stage Profile Table (Rows 16–40)

Aqueous concentrations at every stage, pulled from SOLVER. Columns: - x_Pr, x_Nd, x_Tb, x_Dy — individual element concentrations - x_NdPr (grouped) — Nd + Pr combined (matches screenshot display) - %_NdPr, %_Dy — fraction of total aqueous REE at each stage

Charts

Chart 1 — Aqueous Distributions, Absolute: shows g/L of NdPr and Dy at each stage. Matches the lower-right chart in the reference screenshot.

Chart 2 — Aqueous Distributions, % of Total: shows the composition as a percentage of total REE in the aqueous phase at each stage. Matches the upper-right chart.

Both charts use stages 1–25 on the X-axis. The crossover point between stages 10 and 11 (extraction/scrub boundary) is where separation occurs.

Part 8 — Validation Against Reference Results

With the default inputs, the model should reproduce:

Output	Expected Value
Loaded Organic at stage 10	19.05 g/L
Raffinate Nd conc.	~15.57 g/L
Raffinate purity	100% Nd
Preg Dy conc.	~18.61 g/L
Preg purity	100% Dy
% Feed — Nd	88.59%
% Feed — Dy	11.41%

If values differ, check: 1. All INPUTS values exactly match the reference parameters 2. The SOLVER sheet has no `#DIV/0!` or `#VALUE!` errors (check columns b and c′) 3. Excel calculation mode is set to **Automatic** (Formulas → Calculation Options → Automatic)

Part 9 — How to Use for Design Studies

Changing Operating Conditions

1. Open INPUTS sheet
2. Modify any value in column B (General Inputs) or column C (Feed Concentrations)
3. Results update automatically — no button click required

Understanding Sensitivity

Parameter	Increasing it...
O (organic flow)	Loads more REE onto organic, may overwhelm scrub
S (strip flow)	Improves stripping but dilutes preg; check Preg Conc.
R (reflux)	Higher R → cleaner separation, lower preg flow
HCl Ext	Higher acid → harder to load organic (lower D)
HCl Scr	Higher acid → more aggressive scrub, pushes weakly-loaded REEs to raff
n_ext	More stages → more complete loading (up to equilibrium limit)
n_scr	More stages → higher purity in preg

Changing the Separation Target

To separate a different pair of elements, update the P and Q values in INPUTS for those elements. The model will automatically recalculate. The key is that the two elements you want to separate must have different D values at the operating HCl normalities — expressed as a **separation factor β = D_heavy / D_light**.

Part 10 — Excel Formula Reference

Below is a translation of the key Excel formulas used in the SOLVER sheet, so you can verify or rebuild them if needed.

D coefficient (cell C5, Pr at stage 1)

```
=IF(A5<=INPUTS!$B$5,
    INPUTS!$C$18 * INPUTS!$B$8 ^ INPUTS!$D$18,
    IF(A5<=INPUTS!$B$5+INPUTS!$B$6,
       INPUTS!$C$18 * INPUTS!$B$9 ^ INPUTS!$D$18,
       INPUTS!$C$18 * INPUTS!$B$10 ^ INPUTS!$D$18))
```

Logic: if stage ≤ n_ext → use Ext HCl; elif ≤ n_ext+n_scr → use Scr HCl; else Strip HCl.

b coefficient (cell E5, Pr at stage 1)

```
=IF(A5>INPUTS!$B$5+INPUTS!$B$6,
    -INPUTS!$B$13 - INPUTS!$B$12/C5,
    IF(A5>INPUTS!$B$5,
```

```
        -INPUTS!$B$13 - INPUTS!$B$12*INPUTS!$B$14/C5,
        -INPUTS!$B$13 - (INPUTS!$B$12*INPUTS!$B$14+INPUTS!$B$11)/C5))
```

c′ forward sweep (cell H5, stage 1 — base case)

```
=F5/E5
```

For stage n > 1:

```
=(F{n}) / (E{n} - D{n}*H{n-1})
```

y back substitution (cell J29, stage 25 — last stage)

```
=I29
```

For stage n < 25:

```
=I{n} - H{n}*J{n+1}
```

Aqueous concentration (cell AI5, x_Pr at stage 1)

```
=IF(C5<>0, J5/C5, 0)
```

Guide version 1.0 — Alind SX Steady-State Model, D2EHPA/HCl, TDMA formulation